MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, GASOLINE;

GENERAL SPECIFICATIONS FOR

This amendment forms a part of Military Specification MIL-T-21870, dated 24 May 1966, and is mandatory for use by all Departments and Agencies of the Department of Defense.

Page 2

2.1 Delete the following references: "MIL-T-16165, MIL-E-55301, and MIL-STD-826."

MIL-M-3971 Delete the title and substitute: "Meter, Time Totalizing, Non-Hermetically Sealed, Electric, AC and DC".

Add the following new references in proper sequence:

" O-E-915 - Extinguisher, Fire, Dry-Chemical, (Portable).

QQ-P-416 - Plating, Cadmium (Electrodeposited).

TT-E-489 - Enamel, Alkyd, Gloss, (for Exterior or Interior Surfaces).

VV-K-220 - Kerosene, Deodorized.

MIL-P-514 - Plate, Identification, Instruction and Marking, Blank.

MIL-T-704 - Treatment and Painting of Materiel.

MIL-F-22191 - Films, Transparent, Flexible, Heat Sealed, For Packaging Applications.

MIL-T-43169 - Tool Kit, Master Mechanics.

MIL-W-52574 - Welding and Welding Procedure Requirements For Manufacture of Equipment Utilizing Steels.

MIL-T-83926 - Tire, Pneumatic: Special Purpose.

MIL-STD-461 - Electromagnetic Interference Characteristics

MIL-STD-810 - Environmental Test Methods.

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MIL-STD-1223 - Administrative Wheeled Vehicles, Treatment, Painting, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.

MIL-STD-1400 - Engines, Gasoline Or Diesel, Methods Of Test.

MS-51064 - Pulley, Groove, Engine Accessory, Drive Belts.

MS-51336 - Lunette-Coupler, Drawbar, Ring.

MS-90495 - Plate, Identification, Blank, Mechanical Equipment."

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2.2 Delete address of "The American Society of Mechanical Engineers" and substitute: "345 East 47th Street, New York, New York 10017".

Page 4

3.1 After first sentence insert: "The individual truck requirements shall be as specified herein and in accordance with the applicable specification sheet."

Line 10, delete "in components to affect interchangeability".

3.1.1 Delete and substitute:

"3.1.1 Preproduction testing and approval. Unless otherwise specified (see 6.2), the supplier shall furnish a truck within the time specified in the contract to prove, prior to starting production, that his production facilities and choice of design detail will produce trucks that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish trucks conforming to this specification. The approved preproduction model may be presented for acceptance as the last unit for delivery under the contract after being rehabilitated to a new appearance. This shall include the furnishing of new tires."

Add new paragraph 3.1.1.1 as follows:

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"3.1.1.1 Initial production. When specified (see 6.2), the supplier shall furnish a truck for testing as specified in 4.3.3.4."

Add new paragraph 3.1.1.1.1 as follows:

"3.1.1.1.1 Material identification. At the time the preproduction model or, if none, the first production model is furnished for examination, the supplier shall furnish to the Government in the form of sketches or outline drawings (dimensions not required) identification of high strength constructional steels used in the fabrication of the truck and the recommended welding repair procedures in accordance with MIL-W-52574 for accomplishing repair of such steels by the Military. Identification shall include type of steel with ASTM designation and grade and location by part or assembly. The above information shall be included in the manuals specified in 3.12."

3.1.2.1 Delete and substitute:

"3.1.2.1 Castings. Castings provided on all trucks shall be at least of the quality normally used in commercial practice, free from injurious blowholes, porosity, shrinkage, cracks, or other defects. The use of repaired castings in which the repairs could adversely affect the strength or intended use of the castings shall not be permitted. Sufficient material shall be furnished to permit machining where necessary for leak-free sealing at covers, closure points, and fasteners."

3.1.3 Add at end of paragraph:
"There shall be no exposed bolts, clamps, gages, fittings, lifting parts, or appendages that can be caught or hooked while the truck is working in a confined space. Hose reel, if furnished, is excluded from this requirement."

3.1.3.2 Delete and substitute:

"3.1.3.2 Plating. When corrosion resistance is specified in accordance with 3.6.22 (see 6.2), all threaded fasteners, washers, and cotter pins required to fabricate the truck, except those designed to be in contact with oils in reservoirs and those inside components which are corrosion resisting, shall be cadmium plated in accordance with type II, class 3 of QQ-P-416, or made of an appropriate type of corrosion resisting material."

3.1.3.7 Lines 11 and 12. Delete "Drain plugs ... complete drainage," and substitute:

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"Drain holes with removable drain plugs shall provide for complete drainage without disconnection hose(s)."

3.1.3.7 At the end of the paragraph add the following:

"There shall be no part of the hydraulic system extending below the frame of the truck unless it is properly guarded against damage. When specified (see 6.2), all special hand tools required for maintenance of the truck shall be provided and shipped with each truck procured. Special hand tools are those not listed in MIL-T-43169. Truck shall be designed so as not to require preventive maintenance by the user in less than 100 hours of operation under normal warehouse operations."

Page 6

3.2.1 Delete and substitute:

"3.2.1 Engine. The engine, including all systems, components, accessories, and auxiliaries shall conform to MIL-E-11275, Group II, Liquid Cooled, except as otherwise specified herein. When specified (see 6.2), the engine shall be a standard commercial-industrial type and as described herein."

3.2.1.1 Delete and substitute:

"3.2.1.1 Power and speed rating. When a Qualified Products List (QPL) engine is furnished, the power and speed rating of the engine shall be such that operation of the truck under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by QPL-11275. When a commercial-industrial engine is furnished, the power and speed rating of the engine shall be such that it shall provide sufficient power and torque to operate the truck under any of the operational requirements and conditions specified herein without exceeding the engine manufacturer's published intermittent horsepower rating, or 90 percent of the engine manufacturer's published maximum horsepower rating (whichever is lower) at the applicable speeds."

3.2.1.2 Delete without substitution.

Page 7

3.2.2 Add "The engine accessories for the standard commercial-industrial type and the military engine shall be in accordance with MIL-E-11275, except those accessories specified herein."

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3.2.2.1, 2nd sentence, Delete and substitute: "The choke control shall be located on the instrument panel, frame, or body; it shall be visible to and within easy reach of the operator when he is in operating position."
Delete 3.2.2.3 and 3.2.2.3.1 and substitute the following:
"3.2.2.3 Alternator. An alternator shall be provided. It shall have output characteristics matched to the electrical load required by the equipment as furnished plus the additional reserve capacity to support the complete electrical system and accessories furnished with the vehicle and a 25 percent safety factor capacity. The alternator shall not utilize externally mounted selenium rectifiers. The alternator shall be driven by single or multiple V-belts conforming to MIL-B-11040 to provide full alternator output at all engine speeds without belt slippage. A protective device shall be provided and matched electrically to the alternator so that the vehicle's electrical system including alternator is protected from damage when incorrect polarity is applied or arc welding maintenance repairs are performed anywhere on the equipment. Fuses are not acceptable.

3.2.2.3.1 Alternator regulator. An alternator regulator shall be supplied if required by the alternator and electrical system design. Provision shall be made within the electrical system for:

(a) Alternator voltage regulation.

(b) Current limiting to protect alternator enclosed diodes or the alternator if applicable.

(c) Disconnection of circuit to the alternator rotor when the ignition switch is off.

(d) Actuation of ammeter or charge indicator lamps where applicable.

(e) Field current relay switch to prevent rotor current from passing through the ignition switch where applicable."

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3.2.2.4 Add the following between the first and second sentences:
"A device shall be furnished to automatically disconnect the starter motor when the engine is running."

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3.2.2.5.1 Delete and substitute:
"3.2.2.5.1 Air induction system. The carburetor and intake manifolds shall conform to MIL-E-11275."

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3.2.2.5.2 Delete last sentence and substitute; "Dry type replaceable cartridge type air cleaners are permitted and shall be of the automotive type."

3.2.2.6, 3.2.2.6.1, 3.2.2.6.2, 3.2.2.6.3, 3.2.2.6.4, and 3.2.2.6.5 Delete and substitute:

"3.2.2.6 Engine cooling system. The cooling system shall conform to MIL-E-11275, with the following exceptions and additions. The fan shall discharge away from the operator. Fan belt takeup shall be not less than 3 percent of belt length, plus an allowance for replacement of a new belt without forcing. Where the top of the radiator is lower than the top coolant outlet of the engine, a surge tank may be used. The radiator shall be protected by a heavy grill or the counterweight and shall be mounted to prevent damage from shock or racking. The radiator shall be protected from mud and gravel thrown by the tires. Removable grille or guard specified in MIL-E-11275 is not required."

Page 10

3.2.2.7 Delete and substitute:

"3.2.2.7 Governor. The governor shall be adjusted to limit the maximum speed of the vehicle as specified in 3.6.5. The engine speed shall not exceed the engine manufacturer's recommended continuous duty speed for commercial-industrial engines and shall not exceed the intermittent duty speed as established by QPL-11275 for QPL engines when operated under any condition specified herein."

3.2.2.8, line 3, delete "power-operated" and substitute "Internal Combustion Engine Powered". In line 8 after "...ground level." , insert new sentence "The muffler shall be in accordance with MIL-E-11275."

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3.2.2.10 After "When specified" insert "(see 6.2)".
3.2.3 9th sentence Delete "rate" and substitute "ratio".

3.2.3 9th sentence At end add "when tested in accordance with MIL-STD-268."

3.2.3 After 11th sentence, insert new sentence "When specified (see 6.2), a certificate of compliance may be furnished in place of a test."

3.2.3 Last sentence Delete "(see 6.2)".

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3.2.4.1 After the table add the following:

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"Unless otherwise specified (see 6.2), hose in accordance with SAE 100R7 is acceptable."

Add new paragraph 3.2.4.2 as follows:

"3.2.4.2 Hose reels. When hydraulic hose reels are used on the mast to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges."

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3.3.1 After the second sentence add the following:

"The width of the truck shall not exceed the dimension specified on the applicable Specification Sheet."

3.3.1.2 At the end of the paragraph add: "Top and bottom of fork tips shall be rounded to a minimum radius of 1/8 inch."

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3.3.2.2 After first sentence insert: "Each wheel shall be mounted on wheel studs or cap screws with self-centering tapered ends or on tapered bolt holes which shall provide a positive method of locating the wheel on the hub."

3.3.2.2 Delete fourth and fifth sentences and substitute: "Tire chain clearances for the drive wheels shall be not less than clearances specified in SAE J683."

3.3.2.2.1, delete paragraph in entirety and substitute:

"3.3.2.2.1 Pneumatic tires. The tires shall be new and unused and shall be selected from the sizes listed on MS-16968. The tires shall be pneumatic, high pressure, tube type or tubeless furnished with non-directional tread. Tire loading shall not exceed the values shown on MS-16968 when tested in accordance with MIL-STD-268. When specified (see 6.2), tire design shall comply with MIL-T-83926 or be of commercial tread design for Foreign Object Damage (FOD) Control, subject to approval of the procuring activity."

3.3.2.1.1 Delete paragraph number and substitute 3.3.2.2.2.

At the beginning of 1st sentence insert the following: "When tube type tires are furnished,"

3.3.3 After first sentence, insert the following new sentence: "There shall be a minimum of 7 inches measured vertically between the lowest

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point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion."

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3.3.4.1, line 1. After "...each wheel" insert the following: "or a combination thereof. Brake backing plate shall be integral with its support or shall be attached and restrained by positive means such as welding, dowel pins, tapered dowel, split ring, or similar method to prevent rotation of the brake backing plate during brake operation. Cap-screws and lockwashers installed with or without adhesives and whether tightened into tapped holes or by nuts are not acceptable unless backing plate is also positively restrained.

3.3.4.2, line 10. After "Adjustment" insert the following: "or replacement.

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3.3.5.5, between 2nd and 3rd sentences, insert: "When specified, towing hook shall accept MS-51336 lunette without binding."

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3.3.5.6, 4th sentence. Delete "2 inches" and substitute "3 inches".

3.3.5.7, delete 4th and 5th sentences and substitute: "Unless otherwise specified (see 6.2), the battery shall be charged and dry, and one filling of electrolyte of the required specific gravity shall be shipped with the truck in a separate container."

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3.4.1, at end of paragraph add the following: "The controls shall show no evidence of deformation when the truck is tested as specified in 4.3.3."

3.4.1.5, at end of paragraph add: "A manual throttle as specified in MIL-E-11275 is not required."

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Add new paragraph 3.4.1.8 as follows:

"3.4.1.8 Deadman controls. When specified (see 6.2), deadman controls shall be provided. Upon removal of the operator's weight from the seat, the deadman control shall apply the service brakes. It shall also automatically return the transmission to neutral, if the engine is running and directional control lever is engaged in other than neutral position. The deadman control for the brakes shall be capable of retaining the truck on a

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slope equal to the slope ascension required of the truck. With seat unoccupied, truck shall remain stopped on specified grade without drifting. Visual examination shall determine if control lever is returned to neutral position. Failure of deadman controls to comply with above will be evidence of failure and be cause for rejection of the truck."

Page 21

3.5.1, delete and substitute:

"3.5.1 Lights. Each truck shall be equipped with one sealed beam floodlight of not less than 35 watts, mounted on the left side of the mast. Directional adjustment of this light shall be a minimum of 45 degrees above and below horizontal plane. Horizontal adjustment shall be a minimum of 300 degrees to illuminate front and rear areas. When specified (see 6.2), trucks shall be equipped with an identical floodlight mounted on the right side of the mast. When specified (see 6.2), each truck for Air Force use shall be equipped with four floodlights, two in front and two at rear. Adequate protection against damage by position or guard shall be provided."

3.5.3, line 6. Delete "Fork hanger," and substitute "When specified (see 6.2), fork hanger". At the end of the paragraph, add: "The attachment and its components shall show no evidence of permanent deformation when tested as specified in 4.4.3."

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3.5.4, delete first sentence and substitute: "When specified (see 6.2), removable closed cab shall be furnished."

3.5.4.1, delete and substitute:

"3.5.4.1 Cab heater. "When specified (see 6.2), cab heater shall be furnished. Cab heater shall be automotive kind with windshield defroster."

3.5.5, delete first sentence and substitute: "When specified (see 6.2), the sideshift attachment specified in 3.5.3 shall include a load-leveling device to compensate for irregular surfaces that may cause the vehicle to tilt sidewise."

3.5.7, add at beginning of paragraph: "An overhead guard shall be furnished."

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3.5.8 , line 2, delete "of the type and content prescribed" and substitute "in accordance with O-E-915, type I, class 2, size 2-1/2 with a minimum UL rating of 8B:C.

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Add new paragraph 3.5.9 as follows:

"3.5.9 Weighing device. When specified (see 6.2), the truck shall be equipped with a weighing device, such as the readout of a pressure transducer or of a Bourdon tube dial indicator and shall be connected to and operate from the hydraulic system for raising and lowering the forks. If a dial indicator is furnished, the dial shall be not less than 6 inches in diameter, and the increments shall be marked in 50-pound subdivisions. The dial shall be capable of being rotated 360 degrees by means of an external tare adjusting knob. Bearing points at the dial perimeter shall provide friction to prevent dial rotation owing to shock loads or vibration. The entire indicator case shall be filled with Base Oil C conforming to Federal Specification VV-K-220 to provide movement lubrication and dampening of the tube and pointer oscilation caused by vibration. There shall be sufficient air space left to allow for oil expansion owing to temperature changes. The indicator shall be fitted with a fully adjustable pulsation damper in an easily accessible position on the case. The weighing device shall be manufactured to an accuracy of 5% of full scale reading as demonstrated when tested in accordance with 4.4.5. Weighing device adjusted for 25% tare shall be capable of being overpressured to 150% from true zero without sustaining damage. The indicator shall be within the view of the operator when he is in the operating position. The truck shall meet all requirements of this specification with weighing device installed."

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3.6.8, delete "those" and substitute "that".

3.6.11, at the end of paragraph add the following: "When the truck is unloaded the underclearance of the truck, other than the mast and the axles, shall be not less than 4 inches. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by structural members from striking obstacles."

3.6.12 Add at beginning of paragraph: "The lift assembly shall be capable of holding the rated load at rated maximum lift not less than two minutes with not more than one-half inch of vertical drift and not more than one-fourth of one degree of rotational drift from the vertical."

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3.16.3, delete and substitute "3.6.13".

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3.6.18, delete last sentence.

3.6.19, at end add "When accelerating to maximum speed in reverse direction, steer wheels shall maintain contact with the surface on which the truck is operating."

Add the following new paragraph:

"3.6.22 Resistance to saline atmosphere. When specified (see 6.2), the electrical components shall be designed and installed to withstand the corrosive effects of the salt fog test specified in 4.4.2 without any loss or deterioration of performance."

3.7 Delete and substitute:

"3.7 Workmanship. The truck shall perform any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist."

3.7.1 Delete and substitute:

"3.7.1 Rivet connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately formed and shall have burrs removed. Rivets shall be driven with power tools. Rivet heads shall be full, neatly made, concentric with rivet holes, in full contact with the surface of the member, and shall be in accordance with SAE J492. Excessive upsetting of rivets to fill holes will not be acceptable."

3.7.3, 1st sentence. Delete "punched, drilled, reamed, or cast" and substitute "formed". At the end of the second sentence add "or method". In the third sentence insert "or method" after "device". At the end of the fourth sentence add ", and all other threaded fasteners shall have tightening and thread engagement in accordance with SAE J475."

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Add the following new paragraphs:

"3.7.4 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having corroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly, except in the performance of designed heat treatment, and overheating shall be avoided in accordance with the recommendations of the metal manufacturer. All modular assembly fabrication shall provide for interchangeability of components.

3.7.5 Machine work. All machined parts shall be manufactured to insure accuracy of assembly through the use of correct jigs, fixtures, or tape controlled machines, or any desired combination of these. Like parts shall be interchangeable."

3.8, delete first and second sentences and substitute: "The truck shall be cleaned, treated, and painted in accordance with MIL-T-704. The trucks shall have one coat of metallic base primer and two coats of synthetic enamel."

3.8.1 Delete and substitute:

"3.8.1 Walkway coating. Floor plates and step surfaces shall be coated or matted with type II or III, color 37038, black, of MIL-W-5044 and FED-STD-595."

3.9, 3.9.1, and 3.9.2 Delete in their entirety and substitute:

"3.9 Electromagnetic interference characteristics. Unless otherwise specified (see 6.2), the truck shall be suppressed in accordance with class III D of Table I of MIL-STD-461."

3.10 Delete in entirety and substitute:

"3.10 Identification marking. The truck shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, (MS 90495),

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Composition C, Grade A, Class I, which shall be in a visible and safe location on the instrument panel or fire wall of the truck. The identifying data to be applied to the identification plates shall be as indicated herein and as defined in MIL-STD-130. The truck shall also be equipped with instruction, warning, and caution plates conforming to MIL-P-514, type III, Composition C, Grade A, Class I, prominently located and describing any special or important procedure to be followed in operating, loading, and servicing of the truck or its components. All plates shall be securely attached to the truck with screws, bolts, or rivets, and shall be furnished and mounted by the suppliers."

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Add new paragraphs 3.10.1 and 3.10.2 as follows:

"3.10.1 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition shall show the silhouette of the forklift in transport position indicating the center of gravity and the location and capacity of the lifting attachments. Plate shall be attached by screws, bolts, or rivets in a conspicuous location.

3.10.2 Wheel loading plate. Each forklift shall be equipped with a wheel loading plate conforming to MIL-P-514, type III, composition C, of type 1, grade A, class 1 material. Plate shall be attached by screws, bolts, or rivets in a conspicuous location. As a minimum the plate shall have the following information:

Wheel loading (no load on forks)
Drive wheels (each) pounds
Steer wheels (each) pounds

Wheel loading (rated load on forks)
Drive wheels (each) pounds
Steer wheels (each) pounds"

3.11, delete and substitute:

"3.11 Vehicle marking. The assigned registration number for each truck shall be painted in black enamel in block numbers three (3) inches high on each side of the upright mast. For pneumatic tired trucks the tire pressures for each tire shall be stenciled in black, block numbers one inch high on each side of the truck near the applicable tire. Unless otherwise specified (see 6.2), vehicle markings and their locations for Air Force use shall be in accordance with MIL-STD-1223."

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Add new paragraph 3.11.1 as follows:

"3.11.1 Special Air Force Marking. When specified (see 6.2), trucks procured for the Air Force shall be marked **in black color number 17038** in accordance with Federal Standard No. 595. Marking paint shall be in accordance with TT-E-489. Characters shall be block-type capitals and Arabic numerals. Marking location and information and character size shall be as follows:

Information	Example	Location	Character Size
Capacity	10,000 Lbs. Cap.	On each side of mast	2 inch height
Rider warning	NO RIDERS	On rear of mast	2 inch height
Service designation and Registration number	U.S.AIR FORCE 70E XXX	On each side and rear	1-1/2 inch height
Tire pressure	TP XX	Over each tire	1/2 inch height."

3.12, line 5, between "packaged" and "by" insert "in a bag of transparent material conforming to type II of MIL-F-22191". At the end of paragraph add "(see 6.2)".

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4.1 Delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

4.3.3.2.1 Delete the 1st sentence and substitute: "The preproduction sample truck shall be tested in accordance with 4.4.1, 4.4.2 (if applicable), 4.4.3 (if applicable), 4.4.4, 4.4.5 (if applicable) and MIL-STD-268 to determine compliance with the specification. Forks subjected to 300 percent of rated load shall be inspected for cracks 6 inches each side of heel by magnetic particle or dye penetrant method."

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Add new paragraphs as follows:

"4.3.3.2.1.1 Engine performance (QPL engine). Prior to examination and tests of the preproduction truck, the engine to be installed in the preproduction truck shall be tested in accordance with Test Method Series 2000 of MIL-STD-1400. Non-conformance to 3.2.1 of this specification or failure to meet the requirements of Test Method Series 2000 of MIL-STD-1400 shall constitute failure of this test.

4.3.3.2.1.2 Intake manifold vacuum. During tests specified to meet requirements of 3.6.1, 3.6.5, 3.6.6 and 3.6.19, the engine speed (RPM) and intake manifold vacuum shall be recorded throughout the speed range. The data recorded shall be compared with the data recorded during conduct of the test in 4.3.3.2.1.1. If at any time during the test the vacuum reading falls below that recorded during Test Method Series 2000, MIL-STD-1400 for the applicable speed it shall be considered as exceeding the QPL rating of the engine and shall constitute failure of the test."

4.3.3.2.3 Delete without substitution.

Add new paragraphs as follows:

"4.3.3.4 Initial production inspection. When required, an initial production (IP) truck will be selected at random by the Government from the first 10 trucks being produced from production tooling and shall be examined in accordance with 4.3.3.1, and shall be tested in accordance with 4.3.3.2.1 to determine conformance to the requirements of this specification. When specified (see 6.2), the inspection will be performed by the Government at a site selected by the Government. Acceptance of this truck shall not exclude the remaining trucks from the quality conformance inspection and acceptance provisions specified in Section 4.

4.3.3.4.1 Inspection failure. Failure of the IP truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3.3.4 shall be cause for rejection of the IP truck and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on trucks previously accepted and produced under the contract. Any deviations found as a result of the IP test will be considered prima facie evidence that all trucks accepted prior to the completion of the IP test are similarily deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.3.3.5 Inspection comparison. The Government may select trucks at any time during the contract production period and subject these trucks to the examination and tests specified in 4.3.3.4 to determine whether the quality of the selected truck is equal to the quality standards established during IP inspection. The inspection will be performed by the Government at a site selected by the Government. Trucks will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model(s) and IP trucks.

4.3.3.5.1 Inspection failure. Failure of any truck when inspected as specified in 4.3.3.5 may be considered cause for refusal by the Government to continue acceptance of trucks until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection."

4.3.4 Delete and substitute:

"4.3.4 Examination of preparation for delivery. The trucks will be examined to determine whether preservation, packaging, packing and marking comply with the requirements of MIL-STD-162."

4.4.1 Delete and substitute:

"4.4.1 Electromagnetic interference characteristics. The preproduction model equipped for electromagnetic compatibility in accordance with 3.9 shall be tested as specified in MIL-STD-461. The supplier shall furnish the contracting officer the report of the test required by MIL-STD-461. Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the preproduction truck shall be used as a model for all production trucks."

Add new paragraphs 4.4.2 and 4.4.3 as follows:

"4.4.2 Saline atmosphere. When specified (see 3.6.22 and 6.2), the truck shall be tested in accordance with MIL-STD-810, Method 509, Salt Fog. Individual electrical components as connected to other components may be tested separately instead of with the entire truck. Any evidence of non-compliance with 3.6.22 and MIL-STD-268 shall be cause for rejection of the truck. Pursuant to Method 509, paragraph 4, the following details are designated:

a. Salt fog test failure criteria shall be corrosion of the test items observed in visual inspection or loss or degradation of performance in accordance with MIL-T-21870, paragraph 4.3.3.2.2.

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b. 5 percent.

c. Insofar as possible mount test items so as to present critical surfaces to fog at 30 degrees from vertical.

d. 72 hours.

e. 168 hours at room temperature while loosely covered with transparent film to retard drying.

f. Visual inspection after 24 hours and 72 hours of exposure, but no washing or operation prior to completion of exposure and drying.

4.4.3 Sideshift attachment. When a sideshift attachment is furnished, the following test shall be performed: Operate the sideshift attachment for 100 full cycles at full speed from extreme left to extreme right and back to extreme left per cycle with the rated load on the forks. During the last 50 cycles, the control shall be released abruptly when the load is in approximately mid position between the left and right extremes and then the sideshifter shall be restarted. Upon completion of the 100 cycles, the attachment and its components shall be examined for compliance with 3.5.3. Any evidence of noncompliance shall be cause for rejection of the truck.

Add new paragraphs as follows:

"4.4.4 Starter disconnect. Start the vehicle by energizing the starter switch and keep the starter switch energized a minimum of 10 seconds after the engine starts. Release starter switch for a minimum of 5 seconds and reactivate starter switch while engine is operating. Any evidence of starter engagement while engine is running or not disengaging when engine starts shall be cause for rejection of the truck.

4.4.5 Weighing device. When a weighing device is furnished, it shall be tested for conformance to the requirements of 3.3.5.7.

(a) The first article truck shall be tested as follows: With the truck at rest on a smooth level surface and mast vertical, a calibrated load equal to the rated capacity of the truck plus sero, minus 1 percent shall be lifted 20 inches by the forks five times. Failure of the weighing device to weigh the calibrated load each time within the required accuracy shall be cause for rejection. With truck at rest on a smooth level surface, tied down, pressure relief valve inoperative, and with indicator adjusted for 25 percent tare, a calibrated load equal to 150 percent plus zero minus 1.5 percent of the rated capacity of the truck shall be lifted 20 inches by the forks. Remove the load and disassemble the indicator for inspection. Reassemble the indicator and repeat the

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first test five times with the calibrated load equal to the rated capacity of the truck plus zero, minus 1 percent. Failure of the weighing device to weigh the calibrated load within the required accuracy and repeatability shall be cause for rejection.

(b) The weighing device on each production truck shall be checked once for accuracy at truck rated capacity by use of the calibrated load used in the first test of paragraph (a).

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5.2 Delete and substitute:

"5.2 Marking. When applicable (see 6.2), in addition to any special marking specified in the invitation for bid, marking shall be in accordance with MIL-STD-129."

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6.2(b) Delete and substitute: "(b) Time required for submission of the preproduction model (see 3.1.1)."

6.2(c) Delete and substitute: "(c) When initial production inspection is required (see 3.1.1.1)."

6.2(d) Delete and substitute: "(d) When special tools shall be provided (see 3.1.3.7)."

6.2(e) Delete and substitute: "(e) When a commercial-industrial engine shall be supplied (see 3.2.1)."

6.2(f) Delete and substitute: "(f) When exhaust system shall meet UL-558 Type GS requirements (see 3.2.2.8.1)."

6.2(g) Delete and substitute: "(g) When fungus and moisture resistance required (see 3.2.2.10).

6.2(h) Delete and substitute: "(h) When transmission and drive assembly test is required in lieu of certificate of compliance (see 3.2.3)."

6.2(i) Delete and substitute: "(i) When mechanical transmission is required (see 3.2.3.1)."

6.2(j) Delete and substitute: "(j) When SAE 100R7 hose is not acceptable (see 3.2.4.1)."

6.2(k) Delete and substitute: "(k) When frame racking is not required (see 3.3.1)."

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6.2(l) Delete and substitute: "(l) When tires shall comply with MIL-T-83926 or be commercial FOD control (see 3.3.2.2.1)."

6.2(m) Delete and substitute: "(m) When batteries shall be other than charged and dry with electrolyte (see 3.3.5.7)."

6.2 Add new subparagraphs as follows:

"(n) When deadman controls shall be furnished (see 3.4.1.8).

(o) When additional light(s) is required (see 3.5.1).

(p) When sideshift attachment is required (see 3.5.3).

(q) When fork hanger is required to provide for third fork (see 3.5.3).

(r) When removable closed cab is required (see 3.5.4).

(s) When cab heater is required (see 3.5.4.1).

(t) When load-leveling device is required (see 3.5.5).

(u) When fire extinguisher is required (see 3.5.8).

(v) When weighing device is required (see 3.5.9).

(w) When resistance to saline atmosphere is required (see 3.6.22, 3.1.3.2, and 4.4.2).

(x) Delete and substitute: "(x) Whether other than MIL-STD-461, Table I, Class III D suppression is required (see 3.9)."

(y) When vehicles procured for Air Force shall be marked other than in accordance with MIL-STD-1223.

(z) When special Air Force marking is required.

(aa) When other than one copy of technical manual shall be furnished with each truck (see 3.12).

(bb) When initial production inspection will be performed by the Government (see 4.3.3.4)."

(cc) Level of preservation, packaging and packing required (see 5.1).

(dd) Special marking requirements (see 5.2).

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